

STONE ENERGY INT'L LTD

Production Deployment, Security Hardening & CMS Operations Manual

Architecture

PHP 8.x / MariaDB /
Vanilla JS

Target Environment

cPanel / Apache / Linux

Document Version

2.0 (Production Verified)

Headquarters

Ibadan, Oyo State,
Nigeria

1. Hosting & Server Prerequisites

STONE ENERGY INT'L LTD is built strictly with Vanilla PHP 8.x and MySQL (PDO), designed specifically for standard shared hosting and cPanel environments with zero external framework dependencies.

- **Web Server:** Apache 2.4+ (with `mod_rewrite`, `mod_headers`, `mod_expires`, `mod_deflate` enabled)
- **PHP Version:** PHP 8.1 or PHP 8.2 with standard extensions: `pdo_mysql`, `fileinfo`, `mbstring`, `openssl`, `json`
- **Database:** MySQL 5.7+ or MariaDB 10.4+ with InnoDB engine and `utf8mb4` charset
- **Security:** SSL Certificate (cPanel AutoSSL or Let's Encrypt)

2. Step-by-Step Live Deployment (cPanel)

Step 2.1: Upload and Extract Files

1. Compress all project files into a ZIP archive (excluding local `.git/` directories).
2. Log in to cPanel, open **File Manager**, and navigate to `public_html/` (or designated subdomain).
3. Click **Upload**, upload the ZIP archive, and extract it directly into `public_html/`.
4. Ensure hidden files are visible in cPanel File Manager to confirm `.htaccess` is present.

Step 2.2: Set Linux File Permissions

Set appropriate Linux file permissions to protect sensitive code while allowing uploads and logging:

- **Folders:** `755` (`rwxr-xr-x`)
- **Files:** `644` (`rw-r--r--`)
- **Writable Directories:** `assets/uploads/`, `logs/`, and `config/` must have `755` or `775` permissions.

Step 2.3: Create Database & Import Schema

1. In cPanel, open **MySQL® Databases**.
2. Create a database: e.g., `cpuser_stoneenergy`.

3. Create a user: e.g., `cpuser_stoneadmin` with a strong generated password.
4. Add the user to the database and check **ALL PRIVILEGES**.
5. Open **phpMyAdmin**, select the newly created database, go to the **Import** tab, choose `database/schema.sql` , and click **Import**.

3. Production Environment Configuration (.env)

Create or update the production `.env` file in the web root. Replace placeholder credentials with actual server parameters:

```
APP_NAME="STONE ENERGY INT'L LTD"
APP_ENV=production
APP_DEBUG=false
APP_URL=https://www.stoneenergyintl.com

DB_HOST=localhost
DB_PORT=3306
DB_NAME=cpuser_stoneenergy
DB_USER=cpuser_stoneadmin
DB_PASS=YourStrongDatabasePasswordHere

MAIL_DRIVER=smtp
SMTP_HOST=mail.stoneenergyintl.com
SMTP_PORT=465
SMTP_USER=info@stoneenergyintl.com
SMTP_PASS=YourCorporateEmailPasswordHere
SMTP_ENCRYPTION=ssl
MAIL_FROM_ADDRESS=info@stoneenergyintl.com
MAIL_FROM_NAME="STONE ENERGY INT'L LTD"

SESSION_LIFETIME=7200
LOGIN_MAX_ATTEMPTS=5
LOGIN_LOCKOUT_MINUTES=15
```

CRITICAL SECURITY DIRECTIVE: Always ensure `APP_DEBUG=false` in production. This guarantees that internal server paths, SQL structures, and stack traces are suppressed, redirecting visitors to graceful error pages (404, 500).

4. Post-Deployment Security Hardening

- **Delete or Lock Installer:** Ensure `config/installed.lock` exists. Delete `install.php` from the server so the installation wizard cannot be re-executed.
- **Enforce HTTPS Redirection:** Verify that `.htaccess` redirects all HTTP requests to HTTPS:

```
<IfModule mod_rewrite.c>
  RewriteEngine On
  RewriteCond %{HTTPS} off
  RewriteRule ^(.*)$ https://%{HTTP_HOST}%{REQUEST_URI} [L,R=301]
</IfModule>
```

- **Security Directives in .htaccess:** The provided `.htaccess` denies direct access to `.env`, `.lock`, `.sql`, `.log`, and prevents PHP script execution inside `assets/uploads/`.
- **Rotate Admin Password:** Log into `https://yourdomain.com/admin/login.php` immediately with default credentials (`admin` / `AdminPassword2026!`) and update to a strong custom corporate passphrase.

5. Staff Roles & Access Control Matrix

Administrative Role	Permitted Operations	Target Staff
Super Admin	Full system control, administrative users, backups, audit trails, and site settings.	Managing Director, Chief Information Officer
Admin	RFQs, client messages, products, projects, services, articles, and content settings.	General Manager, Head of Operations
Editor	Create and update product items, project case studies, and blog articles.	Senior Procurement Officer, Project Engineer
Content Manager	Upload site photography, update news articles, edit descriptions.	Corporate Communications Officer

6. Day-to-Day Operations & Commercial Workflow

6.1 Request for Quotation (RFQ) Processing

1. Incoming RFQs appear in `/admin/rfq.php` with a status badge (**NEW**).
2. Click **Review** → to open the full technical specification, customer contact details, item breakdown, and attached tender PDF or drawing.
3. Update the operational status: **NEW** → **UNDER REVIEW** → **QUOTATION PREPARED** → **SENT** → **NEGOTIATION** → **APPROVED** → **COMPLETED** .
4. Check "**Send email notification to client**" to send an automated status email with your notes.
5. Click **Print RFQ Sheet** to generate a print-ready technical tender worksheet for physical procurement files.

6.2 Product Catalog & Technical Datasheet Management

1. Navigate to `/admin/products.php` → Click + **Add New Product**.
2. Enter product specifications separated by double pipes (e.g., `Material: ASTM Carbon Steel | Pressure: ANSI 300# | End Connection: Flanged RF`).
3. Upload the **Primary Product Image** and optional **Additional Gallery Photos**.
4. Upload the **Technical Datasheet (PDF)**. This automatically generates a "Download Technical Datasheet (PDF)" button on the public product page.

6.3 Project Portfolio Case Studies

1. Navigate to `/admin/projects.php` → Click + **Add New Project**.
2. Enter project title, sector category, location, and client name (use `[ADD CLIENT NAME]` if confidential).
3. Enter scope of work bullet points separated by `||` .
4. Upload the **Featured Image** and multiple **Site Gallery Photos**. Visitors can interactively preview photos on the project detail page.

6.4 Site Settings & Corporate Identification

1. Navigate to `/admin/settings.php` .
2. **General & Legal:** Update the CAC registration number and RC number once verified.
3. **Headquarters & Contact:** Verify physical address in Ibadan, telephone hotlines, corporate email, and Google Maps embed code.
4. **WhatsApp Widget:** Enable/disable floating WhatsApp button and customize pre-filled greeting message.
5. **SEO & Analytics:** Input Google Analytics 4 ID (`G-xxxxxxxxxx`) and Google Search Console verification meta tag.

7. Automated Backup & Disaster Recovery

- **Manual Backup:** Go to `/admin/backup.php` → click **Generate & Download SQL Backup Now**.
- **Automated cPanel Daily Cron:** In cPanel → **Cron Jobs**, schedule the following daily command at 02:00 AM:

```
mysqldump -u cpuser_stoneadmin -p'YourPassword' cpuser_stoneenergy | gzip > /home/username/backups/db_
```

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